

Introduction

By now we have an idea of what Blockchain is and how it is implemented in a distributed system. Let us revise the salient features of a blockchain structure that make it stand out from all other data communication models:

1. **Distributed ownership** – By definition, a blockchain is a distributed transactional ledger that is recorded chronologically and publicly. This feature differentiates it from all other existing transaction models. Tampering around is not a possibility in blockchain as all the data is publicly distributed.
2. **Hashing mechanism** – Every block of a blockchain system comprises of a 'hash' which is like its digital fingerprint. This fingerprint gets altered every time a change is made in the dataset of the block. Also, every block has a hash address component that keeps track of the last hash. This is done to maintain a digital continuity in the ledger and to make sure that no block is tampered with. Because if any block is tampered with, its hash will change and thus render the hash address of the next node useless, hence nullifying the entire continual chain. This security measure is one of the main reasons that it has gained such popularity and give a perfect platform for the exchange of cryptocurrency like Bitcoin.
3. **Proof-Of-Work** – Nowadays, we have computers capable of an insane amount of computation in almost no time. Thus, it is possible to bypass the re-hashing mechanism of the Blockchain system by calculating the new possible hash and fooling the next block into believing there has been no tampering. As a failsafe to this loophole, Blockchain implements an elegant solution known as 'proof-of-work' that gives a delay of 10 minutes every time a new hash is created. This assures that the blockchain is truly tamper free. All these features make blockchain the safest and best possible solution for digitisation of data assets and public transaction ledgers.

The internet is evolving into a new technology paradigm based around smart systems, blockchain platforms, and IoT. In the coming years, these technologies will interact and converge in new and unpredictable ways. The convergence of IoT and blockchain will be a major part of this evolution, enabling new automated service systems and business models. Being that as it may, there are still many unresolved issues regarding Blockchain that remain. Internet of Things is the merging of our engineered physical environment with information systems to create smart, adaptive objects and environments. Blockchain is a fast-growing, increasingly complex network